

BEAMOTIL TABLET

DESCRIPTION

A 5.5 mm diameter, round, convex, white tablet.
Each tablet contains Diphenoxylate HCl 2.5 mg, Atropine Sulphate 25 mcg.

INDICATIONS

Diphenoxylate and atropine combination is indicated in adults, as an adjunct to fluid and electrolyte therapy, in the symptomatic treatment of acute and chronic diarrhoea. Beamotil may also be used to reduce the frequency and fluidity of the stools in patients with colostomies or ileostomies.

PHARMACODYNAMICS

Diphenoxylate HCl is a synthetic derivative of pethidine with little or no analgesic activity. It probably acts both locally and centrally to reduce intestinal motility. Atropine is a tertiary amine antimuscarinic alkaloid with both central and peripheral actions. It reduces smooth muscle tone and diminishes gastric and intestinal motility but has little effect on gastric secretion. However, in this preparation, atropine is included in doses below the therapeutic level in an attempt to prevent abuse by deliberate overdosage.

PHARMACOKINETICS

Diphenoxylate HCl is well absorbed from the gastrointestinal tract. It is rapidly and extensively metabolized in the liver principally to diphenoxyl acid (difenoxin) which has antidiarrhoeal activity; other metabolites include hydroxydiphenoxyl acid. It is excreted mainly as metabolites and their conjugates in the faeces; less than 10% is excreted in urine. The half-life of diphenoxylate is about 2.5 hours while that of diphenoxyl acid is about 4.5 hours. It may also be excreted in breast milk.

Atropine is also well absorbed from the gastrointestinal tract. It is rapidly cleared from the blood and is distributed throughout the body and crosses the blood-brain barrier. It is incompletely metabolised in the liver and about 35%-50% of a dose is excreted in the urine as unchanged drug and metabolites. A half-life of about 2.5 hours has been reported. Atropine crosses the placenta and traces appear in milk. The onset of action of Beamotil is about 45-60 minutes while its effects lasts for 3-4 hours.

DOSAGE

Adult

Initial: 2 tablets 3-4 times daily.
Maintenance: 2 tablets once daily as needed.
Usual adult prescribing limit: Up to 8 tablets daily.

Pediatric

Children under 6 years of age: Use is not recommended.
Children from 6-9 years of age : 1 tablet 3 times daily
Children from 9-12 years of age : 1 tablets 4 times daily

SIDE EFFECTS

Beamotil Tablet is well tolerated. Rarely, paralytic ileus or toxic megacolon (bloating, constipation, loss of appetite, severe stomach pain with nausea and vomiting, abdominal distension), and pancreatitis have occurred. Side effects attributed to atropine include mild anticholinergic effects (blurred vision, difficult urination, dryness of skin and mouth, fever), CNS depression (dizziness or lightheadedness, drowsiness, mental depression, restlessness), euphoria, headache, confusion, numbness of hands and feet, and hypersensitivity reactions characterised by angioedema, urticaria, skin rash or itching and swelling of the gums have been reported. Since atropine is present in a subtherapeutic dose, the appearance of these symptoms probably indicates overdosage.

CONTRAINDICATION

Beamotil is contraindicated in patients with hypersensitivity to any of the ingredients. Patients with severe colitis may develop toxic megacolon and atropine's effect on the lower esophageal sphincter may exacerbate reflux. In diarrhoea associated with enterotoxin-producing bacteria and pseudomembranous colitis resulting from treatment with broad-spectrum antibiotics, inhibition of peristalsis may delay the removal of toxins from the colon, thereby prolonging and/or worsening the diarrhoea. Beamotil is also contraindicated in patients with jaundice.

Warning: Not recommended for children under 6 years of age.

PRECAUTIONS

Appropriate fluid and electrolyte should be given to protect against dehydration in all cases of diarrhoea. Oral rehydration therapy which is the use of appropriate fluids including oral rehydration salts remains the most effective treatment for dehydration due to diarrhoea. The intake of as much of these fluids as possible is therefore imperative. Beamotil should be used with caution in patients with hepatic dysfunction as diphenoxylate may precipitate hepatic coma. Hepatic function determinations are recommended at periodic intervals during long-term therapy, especially for patients with hepatic function impairment. Patients with inflammatory bowel disease receiving Beamotil should be carefully observed for signs of toxic megacolon. Drug-induced inhibition of peristalsis may result in fluid detention in the intestine, which may aggravate and mask dehydration and depletion of electrolytes, especially in young children. If severe dehydration or electrolyte imbalance is present, Beamotil should be withheld until appropriate corrective therapy has been initiated; fluid loss may have serious consequences, such as circulatory collapse and renal failure. It should be used with caution in diarrhoea caused by poisoning, until the toxic material has been eliminated from the gastrointestinal tract. Sole treatment with antiperistaltic agent may be inadequate in acute dysentery characterised by bloody stools and elevated temperature; antibiotic therapy may be required. Use of Beamotil in patients with gastrointestinal tract obstruction is cautioned. Diphenoxylate may also increase the risk of respiratory depression in patients with hypothyroidism and respiratory diseases. Although unlikely with usual doses of Beamotil, atropine may aggravate conditions such as acute angle-closure glaucoma, hiatal hernia associated with reflux esophagitis, hypertension, conditions characterised by tachycardia such as hyperthyroidism, heart failure, and cardiac surgery where it may further accelerate the heart rate, intestinal atony of the elderly or debilitated and myasthenia gravis. Patients with overflow incontinence secondary to constipation may be worsened by Beamotil; it may also cause dilation or pseudo-obstruction of the colon. Reduction in tone of urinary bladder may aggravate or lead to complete urinary retention in patients with prostatic hypertrophy, urethral stricture or urinary retention. Elimination of atropine may be decreased in renal function impairment leading to increased risk of side effects. Atropine utilises the same tubular mechanism of excretion as phenolsulfonphthalein (PSP), resulting in decreased urinary excretion of PSP; concurrent use of atropine is not recommended in patients receiving PSP excretion test. Serum amylase values may be increased as a result of spasm of the sphincter of Oddi.



DRUG INTERACTIONS

Concurrent use of Beamotil with addictive medications, especially central nervous system depressants with habituating potential may increase the risk of habituation. Alcohol, barbiturates and other CNS depressants may increase the CNS depressant effects of diphenoxylate. Anticholinergics or other medications with anticholinergic action such as tricyclic antidepressants, amantadine, some antihistamines, butyrophenones and phenothiazines may enhance the effects of atropine during concurrent use; significant interaction is unlikely with usual doses of Beamotil but may occur with its abuse. Monoamine oxidase inhibitors (MAOIs), including furazolidone procarbazine and selegiline may precipitate hypertensive crisis if used concurrently with diphenoxylate. MAOIs may block detoxification of atropine, thus potentiating its action. Administration of naltrexone to a patient physically dependent on opioid drugs, such as diphenoxylate, will precipitate withdrawal symptoms; symptoms may appear within 5 minutes of naltrexone administration, persist for up to 48 hours, and be difficult to reverse. Naltrexone blocks the therapeutic effects of opioids including the antidiarrhoeal effects; also, patients receiving naltrexone should be advised to use alternative antidiarrhoeals when necessary.

OVERDOSAGE

Signs of overdose may be delayed and patients should be observed for at least 48 hours after an overdose. Young children are particularly susceptible to the effects of overdose. Symptoms of overdose include severe anticholinergic effects (continuing blurred vision or changes in near vision; bradycardia followed by tachycardia, with palpitations and arrhythmias, constipation, severe drowsiness, severe dryness of mouth, nose, throat, unusual warmth, dryness and flushing of skin, difficulty in swallowing and talking, photophobia), respiratory depression (severe shortness of breath or troubled breathing), coma, unusual excitement, nervousness, restlessness, irritability, confusion, ataxia, paranoid and psychotic reactions, incoordinations, hallucinations, delirium and seizures. Respiratory depression may occur as late as 12-30 hours after ingestion.

Treatment of Beamotil overdose involves the following:

- Intravenous administration of 0.4 mg (0.01 mg per kg of body weight in children) of narcotic antagonist naloxone, which may be repeated at 2-3 minutes intervals, for respiratory depression.
- Induction of vomiting, or gastric lavage, if vomiting has not occurred.
- Administration of a slurry of 100 grams of activated charcoal, after induction of vomiting or gastric lavage, in non-comatose patients to reduce absorption.
- Supportive therapy such as respiratory support and administration of diazepam to control marked excitement and convulsions.
- Careful monitoring for 48-72 hours.

Up-to-date information on treatment of overdose can be obtained from the National Poison Centre, University Sains Malaysia (Tel: 800-8099)

TOXICOLOGY

Carcinogenicity / Mutagenicity : No information was available.

Pregnancy/Reproduction

Pregnancy: Adequate and well-controlled studies in humans have not been done. Although studies in animals with diphenoxylate and atropine have not shown any evidence of teratogenicity, risk-benefit must be considered since a study in rats showed that maternal weight gain was reduced when diphenoxylate was given at doses of 20 mg per kg per day. Also, at the same dosage, fertility was decreased, and out of 27 matings only 4 rats conceived and bore 25 normal young. Studies in rabbits showed no embryotoxic, teratogenic, or contraceptive effects. (FDA Pregnancy Category C)

Breast-feeding : Problems in humans have not been documented. However, both diphenoxylate and its metabolite, diphenoxyl acid, and atropine are distributed into breast milk.

Pediatrics : Diphenoxylate and atropine combination is not recommended for treatment of diarrhoea in children. Recommended treatment consists of oral rehydration therapy to prevent loss of fluids and electrolytes, nutritional therapy, and if possible, elimination of the underlying cause of diarrhoea. Infants and young children are especially susceptible to toxic effects of atropine. Children may also be more susceptible to the respiratory depressant effects of diphenoxylate.

Geriatrics : No information is available on the relationship of age to the effects of diphenoxylate and atropine in geriatric patients. However, elderly patients may be more susceptible to the respiratory depressant effects of diphenoxylate, and to the mild anticholinergic effects and confusion caused by atropine. In geriatric patients with diarrhoea, caution is recommended because of the risk of fluid and electrolyte loss.

PRESENTATION

Blister pack : 100 x 10's, 10 X 10's

STORAGE CONDITIONS AND USER INSTRUCTIONS

Store in a dry place below 30°C
Protect from light.
Keep out of reach of children.
Jauhi daripada kanak-kanak
Shelf life: Please refer to outer package.
Take medicine with food or meals if gastrointestinal irritation occurs.
Do not take more than the amount prescribed.
Contact your physician if diarrhoea is not controlled within 48 hours and/or fever.
Oral rehydration therapy and dietary treatment of diarrhoea is recommended whenever possible especially in children.
May cause drowsiness.
Avoid alcoholic beverages or CNS depressants.

Route of administration: Oral.

Product Registration Holder:
Duopharma Manufacturing (Bangi) Sdn. Bhd.
Lot No. 2, 4, 6, 8 & 10, Jalan P/7, Section 13, Bangi Industrial Estate,
43650 Bandar Baru Bangi, Selangor, Malaysia.

Manufactured by :
Duopharma Manufacturing (Bangi) Sdn. Bhd.
Lot No. 2 & 4, Jalan P/7, Section 13, Bangi Industrial Estate,
43650 Bandar Baru Bangi, Selangor, Malaysia.



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335mm

185mm

Black 100%